

Project – Drug Safety and Pharmacovigilance Analytics

Context

Before and after a medicine is released to the market, regulatory agencies and pharmaceutical companies continuously monitor its safety. This process is called **pharmacovigilance**.

Pharmacovigilance aims to detect, assess, understand, and prevent adverse effects associated with medicines. Every year, millions of adverse drug event reports are collected from hospitals, pharmacies, healthcare professionals, manufacturers, and regulatory agencies.

These reports are valuable for:

- Detecting potential safety signals.
- Identifying drugs associated with severe adverse reactions.
- Monitoring the evolution of drug safety over time.
- Supporting regulatory decisions.
- Improving patient safety.

Because the volume of reports can become very large, Big Data technologies are increasingly used to analyze pharmacovigilance databases.

In this project, you will develop a **Big Data analytics platform using Apache Spark** to analyze a large collection of drug adverse event reports and extract meaningful safety indicators. The application should be easy to use and provide clear results to support decision-making.

Dataset

- The project is based on a pharmacovigilance dataset containing reports of adverse drug events.
- Each record corresponds to a reported adverse event associated with a particular medication.

- The dataset may originate from public pharmacovigilance repositories, healthcare organizations, or pharmaceutical companies.

Dataset Attributes

Attribute	Description
report_id	Unique identifier of the safety report
report_date	Date when the event was reported
drug_name	Suspected drug
drug_class	Therapeutic category of the drug
adverse_event	Reported adverse effect
severity	Severity level of the event
outcome	Outcome of the event
age_group	Age category of the affected individual
country	Country where the report was submitted
manufacturer	Pharmaceutical company
source_type	Origin of the report
seriousness_score	Numerical seriousness indicator

Example Dataset (see the attached CSV file)

Notes: You are free to enrich the dataset, create additional derived attributes, or reduce the dataset size if necessary for testing purposes. The application must implement at least the minimum requirements described below. Additional features and improvements are encouraged.

Minimum Requirements

Part 1 – Apache Spark

Develop a data analytics application using Apache Spark running in Docker. The application must be implemented using PySpark. Perform analyses such as:

Drug Analysis

- Top reported drugs.
- Number of reports by drug.
- Number of severe reports by drug.

Adverse Event Analysis

- Most frequent adverse events.
- Most severe adverse events.

Temporal Analysis

- Monthly evolution of reports.
- Monthly evolution of severe cases.

Risk Score Computation

Compute a custom risk score for each drug using: Number of reports and Severity levels. Rank drugs according to their estimated risk.

Signal Detection

Identify adverse events disproportionately associated with specific drugs. Example: Which drug is frequently linked to hospitalization?

Part 2 – Data Storage

Store some results in a NoSQL database.

Beyond the minimum requirements, students are encouraged to propose and implement an extension supporting real-time data processing.

Final Presentation

During the final session: An oral presentation: Slides + a live demonstration of the application.

- Maximum duration: 10 minutes.
- Questions may be asked on the course content.
- Each group must submit a PDF report (on Moodle) .